



HOMEWORK

Homework 3: AI Report Validation System — Submission Report

 *Compiled for course submission*

Checklist Mapping (per Reference Docs/HOMEWORK3.md)

Rubric bucket	Points	Where addressed in this file
Writing component (~500 words, your own explanation)	30	Writing component
Git repository links	20	Git repository links
Screenshots / outputs (4–5+)	25	Screenshots / outputs
Documentation	25	Documentation



Git repository links

Base repository: [Agentic-Flight-Report](#) (JdogLloyd1 / main branch).

Resource	Purpose
hw quality control/run_validation.py	Validation system script — calls the validator LLM, writes scores to CSV
hw quality control/validation_criteria.md	Validation criteria / rubric definition (dimensions, scale,

Resource	Purpose
	benchmarks)
hw quality control/experiment_data/validation_scores.csv	Machine-readable validation results for every scored synthesis
hw quality control/run_statistics.py	Statistical comparison (Bartlett, ANOVA / Welch, boxplot export)
hw quality control/run_agent3_experiment.py	Prompt experiment runner (Prompt A / B / C, n runs per prompt)
hw quality control/experiment_data/synthesis_runs/A_001.md	Example synthesized report validated in the experiment (<i>one of thirty; see sibling B_*, C_*</i>)

Pipeline overview (Phases 0–4): `HOMEWORK_PIPELINE.md` .



Screenshots / outputs

1. Validation system in action

```
[hw_flight] Ollama POST start label=chat_text model=gpt-oss:120b-cloud messages=2 tools=False url=https://ollama.com/api/chat
[hw_flight] Ollama POST done label=chat_text http=200 elapsed_s=17.57 tool_calls=0 content_chars=381
[hw_flight] validate row 28: C_008 chars=4339
[hw_flight] agent_run: model=gpt-oss:120b-cloud tools=no task_chars=10583
[hw_flight] chat_text: single Ollama request (no tools)
[hw_flight] Ollama POST start label=chat_text model=gpt-oss:120b-cloud messages=2 tools=False url=https://ollama.com/api/chat
[hw_flight] Ollama POST done label=chat_text http=200 elapsed_s=22.45 tool_calls=0 content_chars=373
[hw_flight] validate row 29: C_009 chars=4717
[hw_flight] agent_run: model=gpt-oss:120b-cloud tools=no task_chars=10961
[hw_flight] chat_text: single Ollama request (no tools)
[hw_flight] Ollama POST start label=chat_text model=gpt-oss:120b-cloud messages=2 tools=False url=https://ollama.com/api/chat
[hw_flight] Ollama POST done label=chat_text http=200 elapsed_s=22.33 tool_calls=0 content_chars=324
[hw_flight] validate row 30: C_010 chars=3505
[hw_flight] agent_run: model=gpt-oss:120b-cloud tools=no task_chars=9749
[hw_flight] chat_text: single Ollama request (no tools)
[hw_flight] Ollama POST start label=chat_text model=gpt-oss:120b-cloud messages=2 tools=False url=https://ollama.com/api/chat
[hw_flight] Ollama POST done label=chat_text http=200 elapsed_s=9.47 tool_calls=0 content_chars=345
[hw_flight] wrote 30 rows to C:\Users\jony1\Documents\GitHub\Agentic-Flight-Report\hw quality control\experiment_data\validation_scores.csv
```

2. Validation output for one evaluated synthesis (scores + rationale)

hw quality control > experiment_data > validation_scores.csv

	A	B	C	D	E	F	G	H	I
1	prom...	run_id	synthesis_path	live_reference_separation	section_covera...	groun...	gap_calibration	conci...	checklist_coverag
2	A	A_001	synthesis_runs/A_001.md	5	5	5	5	4	100
3	A	A_002	synthesis_runs/A_002.md	4	5	5	5	4	100
4	A	A_003	synthesis_runs/A_003.md	5	5	5	5	4	100
5	A	A_004	synthesis_runs/A_004.md	4	5	5	5	4	100
6	A	A_005	synthesis_runs/A_005.md	5	5	5	5	4	100
7	A	A_006	synthesis_runs/A_006.md	5	5	5	5	4	100
8	A	A_007	synthesis_runs/A_007.md	5	5	5	5	4	100
9	A	A_008	synthesis_runs/A_008.md	5	5	5	5	4	100
10	A	A_009	synthesis_runs/A_009.md	5	5	5	5	4	100
11	A	A_010	synthesis_runs/A_010.md	5	5	5	5	4	100
12	B	B_001	synthesis_runs/B_001.md	5	5	5	5	4	85
13	B	B_002	synthesis_runs/B_002.md	5	5	5	5	4	90
14	B	B_003	synthesis_runs/B_003.md	5	5	5	5	4	100
15	B	B_004	synthesis_runs/B_004.md	5	5	5	5	4	100
16	B	B_005	synthesis_runs/B_005.md	5	5	5	5	4	45
17	B	B_006	synthesis_runs/B_006.md	5	5	5	5	5	100
18	B	B_007	synthesis_runs/B_007.md	5	5	5	5	5	100
19	B	B_008	synthesis_runs/B_008.md	5	5	5	5	4	90
20	B	B_009	synthesis_runs/B_009.md	5	5	5	5	4	70
21	B	B_010	synthesis_runs/B_010.md	5	5	5	5	4	100
22	C	C_001	synthesis_runs/C_001.md	5	5	5	5	4	100
23	C	C_002	synthesis_runs/C_002.md	5	5	5	5	4	100
24	C	C_003	synthesis_runs/C_003.md	5	5	5	5	4	85
25	C_004	synthesis_runs/C_004.md	5	5	5	5	5	4	100

3. Validation criteria / rubric (text criteria)

Course rubric allows a rubric screenshot *when the criteria are visual*. Here the rubric lives in `validation_criteria.md` as structured tables rather than GUI tool output. **The canonical criteria table for grading is reproduced under [Documentation](#) → [Validation criteria table](#)** so every dimension remains traceable without a redundant figure.

(Optional for your Canvas .docx : insert a screenshot of `validation_criteria.md` rendered in your editor or browser if your grader prefers a photographic rubric artifact.)

4. Statistical analysis (terminal output — Bartlett + ANOVA)

```
=== Run overview ===
Source: C:\Users\jonyl\Documents\GitHub\Agentic-Flight-Report\hw quality control\experiment_data\validation_scores.csv
Rows: 30 | prompt_id: A, B, C (3 group(s))

Distinct run_id per prompt_id:
prompt_id
A      10
B      10
C      10

overall_score — all rows:
count    30.0000
mean      4.5600
std       0.2647
min       4.0000
25%      4.4000
50%      4.6000
75%      4.8000
max       4.8000

Dimension means (1-5, all rows):
live_reference_separation  4.7667
section_coverage          4.5000
grounding                 4.9667
gap_calibration            4.8333
concision                 3.7333

checklist_coverage_pct — all rows:
count    30.00
mean     98.67
std       7.30
min       60.00
25%      100.00
50%      100.00
75%      100.00
max      100.00

grounding_gate:
grounding_gate
True      30

validator_model: gpt-oss:120b-cloud

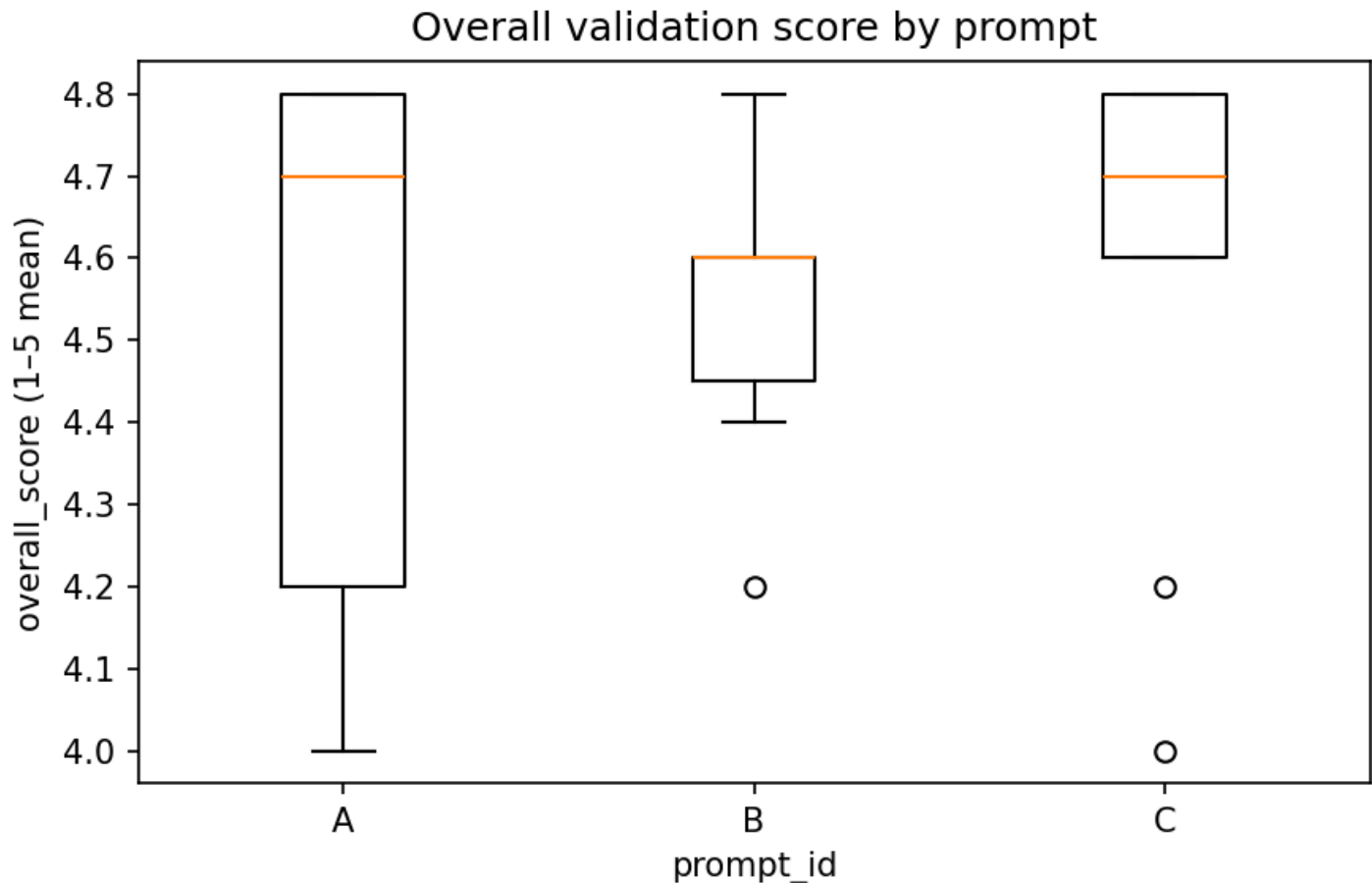
overall_score by prompt_id (mean, std, count):
      mean    std  count
prompt_id
A      4.52  0.3293    10
B      4.56  0.1838    10
C      4.60  0.2828    10
=== End overview ===

Bartlett homogeneity p=0.245906

ANOVA (equal variances):
      Source  ddof1  ddof2    F    p_unc    np2
0 prompt_id    2     27  0.216  0.807114  0.015748

Wrote plot: C:\Users\jonyl\Documents\GitHub\Agentic-Flight-Report\hw quality control\experiment_data\overall_score_by_prompt.png
exit code: 0
```

5. Comparison of scores across prompts (box plot)



Writing component

I built a quality control agent that evaluates the quality of a data reporter module from my Agentic Flight Report pipeline. In the agentic pipeline, Agent 3 aggregates API tool calls from Agent 1 and RAG information from Agent 2 to craft a coherent picture of airspace risk to the flight in question; Agent 4 then takes this output and condenses it for the customer. I tested 3 different prompts for Agent 3 with a fixed Agent 1/Agent 2 input to isolate the impacts of prompting. The results of each prompt were evaluated on a strict 1-5 scale system for different validation criteria, combined with a checklist coverage percentage and a Boolean grounding gate. Instead of reusing the class lab's generic essay quality Likert scales, I separate scoring by sector: live/reference separation, honest gap-calibration vs empty tools, grounding to JSON, capped concision penalties. The full breakdown can be viewed in `validation_criteria.md`.

For the experiment I pulled one frozen upstream snapshot off the manifest

(`experiment_data/run_manifest.json`): AA 849 DFW→BOS, 2026-04-27. Prompt flavors were A*

(baseline airspace), **B** (exec-brief vibe), **C** (lead with uncertainty). Each prompt was run $n = 10$ times, resulting in 30 scored rows overall (`A_001 ... C_010`).

I used a modified lab statistics script titled `run_statistics.py` . That script looked at `overall_score` — the mean of the five core 1–5 scores in each CSV row—and asks whether **A/B/C** could plausibly differ. It prints the verbal analysis, runs Bartlett to sanity-check variance across groups, and if $\alpha = 0.05$, it runs a standard one-way ANOVA test on $H_0: \mu_A = \mu_B = \mu_C$. If the spreads did not meet the threshold, the script automatically switches to Welch ANOVA instead. A `prompt_id` box plot saved as `overall_score_by_prompt.png` was also generated to eyeball overlap and weird outliers.

In my actual results, Bartlett came back around $p \approx 0.246$, so the pooled ANOVA path was run instead of Welch. The omnibus test was pretty flat: $F_{2,27} \approx 0.216$, $p \approx 0.807$, partial $\eta^2 \approx 0.016$ —so none of the prompts stood out on their own as clearly better solution based on `overall_score` . The sample means hug each other ($C \approx 4.60$, $B \approx 4.56$, $A \approx 4.52$). The closeness in scores may be due to only 10 samples per prompt and Agent 3 not tightly bound enough to phrase things differently even when the upstream JSON is fixed, meaning more drastically different prompt wording may be necessary to demonstrate a validation difference.

I tweaked the rubric a few times early on — the validator was initially handing out closely spaced, very high scores to almost everything. I added more granular details for scoring levels and verbiage such as "if you're unsure, score lower." Changes resulted in some more spread observed in the scoring, but the validator still struggled to give scores lower than a 3, if any. Improvement efforts for a production effort would first work on tightening the scoring system to grade more critically.

Documentation

System design summary

Agent 3 writes syntheses (`experiment_data/synthesis_runs/*.md`) from frozen Agent 1 + Agent 2 artifacts. `run_validation.py` submits each Markdown file to a validator model (role `VALIDATOR_ROLE` in that script — see repo link) structured to emit JSON dimension scores aligned with `validation_criteria.md` . Scores accumulate in `validation_scores.csv` . `run_statistics.py` reads the CSV and runs **Bartlett** then **ANOVA** (or Welch ANOVA if variances diverge).

Validation criteria table

Dimension	Description	Scale	Benchmark
live_reference_separation	Clarity distinguishing LIVE operational snapshot vs FAA reference/context	Strict 1–5	Operational ≥4 only if tagging is coherent end-to-end
section_coverage	Required operational themes rendered with specifics (NAS, delays, hubs, gaps) — not vague mentions	1–5	≥4 demands concrete grounding in inputs
grounding	Numeric and factual alignment with LIVE JSON / Agent 2 reference	1–5	Penalize invented statistics or misplaced confidence
gap_calibration	Honesty about sparse / failed tool payloads	1–5	≤3 if missing data is falsely glossed over
concision	No boilerplate stuffing; earns length operationally	1–5	Penalize repeating reference slabs
checklist_coverage_pct	Structural completion of substantive ## sections	0–100 (nearest 5)	Separate from LAB-style abstract Likerts
grounding_gate	Hard contradiction check vs LIVE facts	true/false	Instant fail flag if breached

Contrast with LAB. The LAB’s generic textual Likerts map general attributes (tone, neutrality, completeness *in the abstract*) *without* enforcing domain-specific fidelity to cockpit-style JSON ingestion. This homework couples rubric anchors to explicit **(live)/(reference) fidelity**, checklist coverage tooling, and a **deterministic-derived overall_score** (mean of the five core Likerts) for omnibus inference.

Experimental design snapshot

Item	Specification
Factors	<code>prompt_id</code> $\in \{ \text{A, B, C} \}$ (three distinct synthesizer prompting strategies)
Runs per prompt	10 scored syntheses each $\rightarrow$ 30 rows in <code>validation_scores.csv</code>
Dependent variable (primary analysis)	<code>overall_score</code> = mean(dimensions 1-5) on [1.0 , 5.0]
Blocking / controls	Frozen Agent 1 + Agent 2 bundle per pilot flight (see manifest)
Validator model (this batch)	<code>gpt-oss:120b-cloud</code> (column logged in CSV)

Statistical analysis narrative

Hypotheses.

- H_0 : $\mu_A = \mu_B = \mu_C$ (equal population means on `overall_score`)
- H_a : at least two prompt populations differ

Procedure. Bartlett $p > 0.05 \rightarrow$ proceed with classical **one-way ANOVA** via `penguin / scipy` .

Empirical realization (frozen experiment log).

Quantity	Value
Bartlett p	≈ 0.246
ANOVA $F(2, 27)$	≈ 0.216
ANOVA p -value	≈ 0.807

Interpretation. Fail to reject H_0 . Prompt choice did **not** produce a statistically detectable shift in aggregated validation quality across this modest sample. Do not claim one prompt “wins decisively”; cite the negligible effect size $\eta_p^2 \approx 0.016$. For coursework science practice, optionally discuss whether **larger n** , narrower prompts, multi-flight blocking, or per-dimension contrasts would sharpen power — but absent those extensions, inference stops at an **equivocal omnibus** result.

Technical details

Detail	Guidance
Python env	hw quality control/standalone/.venv (install from standalone/requirements.txt ; includes pandas , pingouin , scipy , matplotlib)
Secrets	Populate standalone/.env (OLLAMA_API_KEY , optional HW_FLIGHT_VALIDATION_MODEL , timeouts, hosts)
Artifact layout	experiment_data/agent1_live.txt , agent2_reference.txt , synthesis_runs/*.md , validation_scores.csv , overall_score_by_prompt.png
Prompt sources	agent3_prompts.py (B, C definitions) plus baseline A in hw_flight.agents.airspace_synthesizer

Usage instructions (minimal)

Executed from `hw quality control/` :

1. **Freeze upstream inputs** (*if refreshing*) — `standalone/run_smoke_agents_12.py` (see README in `standalone/`).
2. **Run prompt experiment:**
`standalone\.venv\Scripts\python.exe run_agent3_experiment.py --n 10 .`
3. **Validate:** same interpreter → `run_validation.py` .
4. **Statistics:** `run_statistics.py` → prints omnibus tables and writes `experiment_data/overall_score_by_prompt.png` .

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